INTERNATIONAL ROADMAP FOR DEVICES AND SYSTEMS™

2024 EDITION

LITHOGRAPHY & PATTERNING

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LITHOGRAPHY AND PATTERNING

1. EXECUTIVE SUMMARY

Lithography and patterning technology have continued to advance but still face many challenges. EUV exposure tools with a numerical aperture (NA) of 0.55 started being delivered to customers at the end of 2023. First production use of such tools is projected for 2027. The availability of both multiple patterning and high NA EUV enables high resolution patterning that already can meet resolution requirements through 2029. Resolution should be extendable to meet requirements through 2033, where minimum metal half-pitch is expected to be 7 nm, and the minimum via pitch is expected to be 31 nm. However, achieving this resolution does not mean lithography requirements are satisfied. Improvements in control of stochastic issues such as defects and line edge roughness are substantial current challenges. Advances in light sources, in mask design, in mask materials, in photoresists, and in ancillary materials such as underlayers will be needed along with improvements in defectivity. All of these will be needed to accommodate the limited depth-of-focus of high NA imaging. Another issue with high NA EUV is the field size, which is half that of the 0.33 NA tools currently used in production. High performance systems that have used large chips in the past will have options to keep improving performance. Two or more exposures for different parts of a chip can be used with the patterns stitched together, or can be used packaging to connect smaller chips and provide high performance overall. Both options have their challenges. Packaging connection pitches will need to shrink to provide improved connectivity as shown below, but larger chips may still be made using stitching, with its attendant challenges. As an alternative, larger format EUV masks are being considered for enabling the patterning of large chips with single exposures.

Starting in 2025, there will be metal lines and spaces patterned on the back of the chip to enable smaller logic unit cells. This will start with four backside layers of metal using 193 nm lithography that provide power distribution and I/O. In 2027 the backside metal will start being patterned with EUV and include more chip functions. 3D logic devices with one or more layers of devices stacked on top of each other are projected to be introduced in 2035 and after that there is little change in minimum chip dimensions.

In the long term, there are some possible new technologies being evaluated to meet these requirements. EUV phase shifting masks, curvilinear mask designs, free electron lasers as a possible light sources, larger mask sizes, directed self-assembly for pattern rectification, and new device designs and process integration are all being investigated. Hyper NA EUV tools with NA's above 0.75 are being considered to improve process windows. New resists based on metals are being actively investigated. Metal resist has the potential to enable thinner films that still absorb sufficient light and provide usable etch resistance. Shorter wavelengths than the current 13.5 nm are also being investigated but their use is farther off than improved NA. Sustainability is also a long-term challenge. Improved materials and reduced power and water usage will be needed.

2. OVERVIEW

Historically, improvements in lithography have been key enablers to make improved chip technologies. The International Roadmap for Devices and Systems (IRDS) Lithography roadmap predicts how current patterning capability can support future chip generations and where challenges and improvements are needed. It gives a roadmap of expected key lithography requirements. It is intended to be used by semiconductor industry participants, by industry analysts, and by researchers who want or need to know how the industry will evolve in the future and what challenges need to be addressed.

This document is the 2024 Lithography and Patterning roadmap. There is an Excel file goes with this document that contains the tables here along with notes. This document and the associated tables cover the years from 2024 to 2039. The tables are projections of what requirements are necessary to maintain industry progress. They are color coded to indicate whether technology is known or unknown that will meet these requirements. There are also difficult challenges, potential solutions and long-term challenges. Different challenges are discussed in detail below.

2.1. DEVELOPMENT OF ROADMAP

This roadmap was developed through consultation with an international team of lithography and patterning experts and review of publicly available literature and other available documents. The current contributing membership is shown in the acknowledgments. Contributing members come from Asia, Europe, and the United States and represent semiconductor, equipment, and material manufacturers, as well as research institutes. Without their expertise and contributions, this roadmap would not exist. The IRDS overall roadmap and the More Moore focus team provide the device roadmap from which lithography requirements are derived. Using input from the Lithography + Patterning focus team members the key options, their timing and their key challenges are developed. Input and joint work with other IRDS IFTs was also necessary, particularly, More Moore,

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Yield Enhancement, Packaging Integration, Environmental, Health, and Safety, and Metrology. The table and this document underwent internal review by the team and by the overall IRDS before publication.

2.2. Lithography Drivers

This roadmap projects future patterning needs and where the key challenges are. In the past, both logic and memory devices have driven improvements in patterning technology. Currently, high-performance logic devices are leading the way to smaller critical dimensions and driving the introduction of novel patterning technology. New DRAM devices are continuing to be introduced with smaller critical dimensions (CDs), but trail logic in smallest resolution. Flash memory is scaling using 3D structures that have relatively large CDs and does not need higher resolution patterning to make progress. Thus, most of the challenges and solutions are discussed below in terms of logic technology challenges.

2.3. Trends since 2022 Lithography Roadmap

EUV lithography, which uses 13.5 nm wavelength light for imaging, is well established. Advanced logic products are dependent on it and DRAM makers have started using EUV lithography in high volume manufacturing. It has become clear that EUV single patterning will not completely replace multiple patterning. ArF immersion multiple patterning will continue to be used and there will be use of multiple patterning, depending on level-specific details, yield, and cost. EUV resolution will improve with the introduction of 0.55NA tools and further improvements in EUV capability are needed. The trend to more three-dimensional structures continues, although actual stacking of logic devices is now projected in 2033 instead of 2031. Backside routing will be introduced in 2025, and the backside structures will become more complex and three dimensional every node. Yield and chip size are also areas of challenge. High performance packaging will be needed to make high performance systems and now holds lithography challenges as described below. Finally, environmental challenges in energy usage and in chemical usage related to lithography and patterning will have to be addressed.

3. Lithography requirements

3.1. Introduction

High performance logic scaling is continuing to drive increases in device density. Through 2033, advanced logic will continue to require the smallest resolution of any chip type. Whether DRAM minimum dimensions decrease below logic's after that depends on the path taken to increase DRAM bits per area. After 2033 DRAMs could either scale dimensions smaller or change architecture to keep improving density. Which option will be chosen is unknown, so the table currently shows static dimensions for DRAMs, but these are very tentative values. The projected increases in device density will be driven not only by pitch scaling, but also by design technology co-optimization (DTCO), increased use of 3D structures for devices, and advanced packaging. Details of projected device scaling are shown in the IRDS More Moore roadmap.[1] These additional factors enable continued performance increases despite slower reduction in critical dimensions than projected in the 2022 roadmap. Despite the rate of pitch reduction being slower, the smallest projected half-pitch now reaches 7 nm rather than 8 nm. These cells are color coded yellow in the table, meaning there is technology available, but not yet demonstrated in manufacturing. High NA EUV may have stochastic issues at this half-pitch, but 0.33 NA multiple patterning should be capable of this pitch assuming continued use of Manhattan layouts. Curvilinear layouts, if adopted, could be more challenging to pattern with multiple patterning. Better EUV patterning at low k_1 will be needed and this is a major challenge.

Improving stochastic effects such as line edge roughness (LER) and defects, improved critical dimension uniformity, and improving edge placement error (EPE) will be more challenging than resolution. How to minimize stochastic effect without unacceptable increase in dose-to-print are the subject of much industry research and development. Better photoresists, modified patterning processes, improved exposure tools and masks will be needed. These trends combine to produce the lithography requirements roadmap shown below. The logic technology naming in the second line shows the gate pitch, the metal pitch and the number of devices tiers in abbreviated fashion. For example, G42M14T2 means that the minimum gate pitch for devices is 42nm, the minimum metal pitch is 14nm and there are two tiers of logic devices with these pitches.

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Table LITH-1 Lithography Technology Requirements

Parameter	2024	2025	2027	2029	2031	2033	2035	2037	2039
MPU / Logic									
Logic device technology naming [F]	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Logic Industry "Node Range" Labeling	"3nm" Enhanced	"2nm"	"1.4nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Logic device structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
MPU/ASIC Minimum Metal ½ pitch (nm)	11.5	11.0	11.0	10.0	9.0	8.0	7.0	7.0	7.0
Metal CD control (3 sigma) (nm) [B]	1.7	1.7	1.7	1.5	1.4	1.2	1.1	1.1	1.1
Contacted poly half pitch (nm)	24	24	24	23	22	22	21	21	21
Physical Gate Length for HP Logic (nm)	16	14	12	12	12	10	10	10	10
Gate LWR (nm) [C]		1.4	1.2	1.2	1.2	1.0	1.0	1.0	1.0
Gate CD control (3 sigma) (nm) [B]	1.6	1.4	1.2	1.2	1.2	1.0	1.0	1.0	1.0
Overlay (3 sigma) (nm) [A]	2.3	2.2	2.2	2.0	1.8	1.6	1.4	1.4	1.4
Metal CDU (nm)	1.8	1.7	1.7	1.5	1.4	1.2	1.1	1.1	1.1
Metal LER (nm) [C]	1.3	1.2	1.2	1.1	1.0	0.8	0.7	0.7	0.7
MPU/ASIC finFET fin minimum 1/2 pitch (nm)	12								
FinFET Fin width (nm)	5.0								
Fin CD control (3 sigma) (nm) [B]	0.50								
FIN LER (nm) [C]	0.35								
Lateral finFET/Gate All Around (LGAA) pitch	24.0	26.0	24.0	24.0	22.0	22.0	22.0	22.0	22.0
LGAA minimum width	7.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Fin/GAA/CFET CDU (nm) - 10%	0.5	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.6
GAA LWR (nm) [C]	0.5	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.6
MPU/ASIC minimum contact hole pitch (nm)	48	48	48	46	44	44	42	42	42
Contact CD (nm) - finFET, LGAA	20	22	26	24	24	26	24	24	24
MPU/ASIC minimum Via pitch (nm) [E]	47	45	45	45	40	36	31	31	31
MPU/ASIC Via half pitch	23	22	22	22	20	18	16	16	16
MPU/ASIC minimum contact hole or via CDU (nm)	3.0	3.3	3.3	3.4	3.0	2.7	2.3	2.3	2.3
Backside Lithography									
Number of backside metals	-	4	4	5	5	6	6	6	6
Backside/via contact pitch (nm)		120	32	27	21	19	16	16	16
First backside metal pitch (nm)		192	32	27	21	19	16	16	16
Backside functions		PDN, IO	PDN, IO, Clock	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus
DRAM									
Dram Minimum 1/2 pitch (nm)	15.5	13.0	14.0	11.5	10.0	7.0	7.0	7.0	7.0
CD control (3 sigma) (nm) [B]	1.6	1.3	1.4	1.2	1.0	0.7	0.7	0.7	0.7
Mininum contact/via after etch (nm) [D]	16	13	14	11.5	10.0	7.0	7.0	7.0	7.0
Minimum contact/via pitch(nm)[D]	47	39	42	35	30	21	21	21	21
Overlay (3 sigma) (nm) [A]	3.1	2.6	2.8	2.3	2.0	1.4	1.4	1.4	1.4
Other Memory									
Other memory ciritical dimensions and patterning specifications are large compared to logic or DRAM									
Chip size (mm^2)									
Maximum exposure field width (mm) [E]		26	26	26	26	26	26	26	26
Maximum exposure field length, i.e. scanning direction (mm) [E]		16.5	16.5	16.5/33	16.5/33	16.5/33	16.5/33	16.5/33	16.5/33
Maximum field area printed by exposure tool (mm^2)		429	429	429/858	429/858	429/858	429/858	429/858	429/858

4. EUV CHALLENGES

4.1. HIGH NA EUV KEY CHALLENGES

Lithographic technologies have long been extended by increases of numerical aperture, and prototype high-NA EUV lithography tools have been fabricated by ASML and were installed for early use by chip makers in 2024. Their 0.55 NA optics (made by Carl Zeiss) compared to existing 0.33 NA tools, provides for higher resolution, which is expected to enable ~16 nm minimum pitch (single patterning). In these tools, the lens reduction in the scan direction was increased to 8x to maintain the reflectivity at the mask by reducing the angle of incidence of illumination. The reduction ratio in the lens slit direction is maintained at 4x. The result is a maximum exposure field size on the wafer of 26 mm x 16.5 mm, which is half the current scanner field length with the current mask size. Scanner throughput can be maintained with two passes by increasing the wafer and reticle stage velocities and accelerations. One solution for chip sizes larger than a half-field is stitching two separate exposures. This requires tight overlay and CD control at the boundary unless an exclusion area is defined.[2] Stitching can also be done through direct write methods. This would require an additional exposure tool and process step. The exclusion area must still have some features for pattern density uniformity for etch and CMP. With two masks/wafer, there will be some loss of throughput for exchanging reticles. Another solution could be larger masks to support the full 33 mm scan with the larger reduction ratio. Currently larger masks are being considered. Making a larger mask requires no fundamental changes in mask making technology but will require tool and infrastructure changes. The increase in mask size would also enable higher throughputs for the high-NA EUV lithography tools. We estimate that the earliest a larger mask size could be introduced is 2028.

High-NA EUV lithography has additional challenges associated with resist materials, mask materials, and process control. Inherent with the high NA is a smaller depth-of-focus which drives thinner resists and tighter wafer flatness control. Thin resists

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drive metrology challenges as addressed in the Metrology chapter. New mask absorber materials have been introduced for 0.33 NA and would extend to 0.55 NA to address mask 3D effects. With the tighter pitches possible with 0.55 NA, the edge placement error control becomes incrementally more difficult, which drives computational lithography and process control. Stochastic defects are addressed in the next section. Table LITH-1A addresses some of the key challenges of 0.55 NA EUV lithography.

Going forward, the EUV tools (all NA) will have to become more energy and resource efficient. Decreasing the energy required to expose each wafer can be accomplished by improved efficiency of generating and transmitting EUV light to the wafer, improved idle and sleep modes, and higher throughputs. Other resource savings options include hydrogen recycling and higher temperature cooling water.

Table LITH-2 The Key Challenges of A) High-NA (0.55) and B) Hyper-NA (>0.75) EUV Lithography

General issues	**Issues for high-NA EUV lithography**
Maximum utilization of optical resolution	Resists meeting resolution requirements, with low levels of defects from stochastic phenomena and pattern collapse and acceptable photo speed
	Light sources that can support photon shot noise and productivity requirements
	Resolution enhancement and computational lithography capabilities to address mask 3D effects, dispersion in attPSM absorbers, and polychromatic illumination
Die cost	Cost of high-NA EUV lithography
Process complexity	Solutions for large dies
Process control	Solutions for meeting small depths-of-focus at 0.55 NA, LER < 1 nm, and overlay ≤ 1.6 nm
Masks	Mask making and metrology infrastructure, including new absorber materials
	Additional Issues for hyper-NA EUV lithography
Maximum utilization of optical resolution	Polarized light sources that can support photon shot noise, short ranges for photoelectrons and secondary electrons, and productivity requirements
Die cost	Cost of hyper-NA EUV lithography
Process control	Solutions for meeting small depths-of-focus at ≥ 0.75 NA and LER < 1 nm, and overlay ≤ 1.6 nm

Longer term, 13.5 nm wavelength EUV could be further extended by developing tools with an NA higher than 0.55, and hyper-NA systems (NA ≥ 0.75) are under consideration to address <16 nm pitches as may be required after 2035. A hyper-NA tool would use much of the infrastructure already in place for current EUV lithography by reusing the basic source, mask, and optics technology. The higher-NA would inherently drive larger optical elements which require incremental improvements in optics polishing, coating, and metrology technology. Polarized light could improve image-quality for some structures, and enable the full potential of EUV optics for NA ≥ 0.75. Since current LPP EUV light sources produce unpolarized light, new technology will need to be introduced for producing polarized light. Using polarizers would reduce the light transmitted to wafer by over a factor of 2, causing a significant reduction in productivity. Alternatively, free-electron lasers, which produce polarized light, could be

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used as sources of EUV light. The depth-of-focus will also be very small for hyper-NA imaging, and this would drive resist thicknesses even smaller and flatness requirements even tighter. Table LITH-1B addresses some of the key challenges of hyper-NA EUV lithography

Shorter wavelengths (6 nm < λ < 7 nm) are also being assessed – sometimes referred to as Beyond EUV (BEUV). Changing to a shorter wavelength would be a much bigger change than increased NA at 13.5 nm wavelength, although shorter wavelength lithography would have larger depths-of-focus than hyper-NA at λ = 13.5 nm for the same resolution. The current Sn based LPP source would need to be replaced by a new light source with different target materials or with a free electron laser (FEL). The existing Mo/Si multilayer reflective coatings would have to be replaced by new materials. Resists with sufficient absorption at the shorter wavelength would have to be developed. Maintaining the same level of photon shot noise at the shorter wavelength would require a doubling of light energy delivered to the wafer while maintaining the efficiency of the optical system (reflectivity of the mirrors and the bandwidth) would become more challenging. The higher delivered dose coupled with the optical system efficiency would increase optics, mask, and wafer heating. Currently, known multilayers with high reflectivity at λ = 6.7 nm have angular bandwidths that would limit resolution to ≥ 6 nm (k_1 = 0.3) at 8x lens reduction, [3 4] larger than could be achieved with 0.77 NA at λ = 13.5 nm.

Table LITH-3 The Key Challenges of shorter wavelength (6-7 nm) EUV Lithography

High-brightness standalone light sources for metrology at 6-7 nm wavelength
High-reflectivity mask multilayers optical coatings and absorbers for 6-7 nm wavelength with sufficient angular bandwidth
Solutions for optics and mask heating
Resists with high-absorption at 6-7 nm wavelength
Photon shot noise

4.2. Mask Challenges

High-NA EUVL presents a new set of technical and economic challenges on the photomask. Its initial implementation uses current EUV mask form factors that give half the exposure field size compared to low NA tools (26 mm x 16.5 mm compared to 26 mm x 33 mm). A major challenge in this phase will be with very large dies, which will require field stitching. However, this currently represents only a few percentages of advanced logic devices. Additionally, a report at Photomask Japan 2024 by IMEC and ASML showed that the stitching accuracy does not seem to be a major issue, with actual accuracy measured at about a third of the required accuracy. The required black border placement accuracy is <30 nm@mask. Actual accuracy was 8~12 nm. Other stitching accuracy had looser requirements.

The implementation of larger mask formats (6" x 11.2" x 0.25") to accommodate 26 mm x 33mm exposure fields appears to be primarily a cost issue, rather than a technical issue. Large format EUV blanks may increase mask cost substantially because of investment for infrastructures and a limited customer set. We need industry wide consensus and cost mitigation to facilitate an affordable path.

New materials for mask blanks, including low-n/low-k and low-n/middle-k absorbers (where n and k are the real and imaginary parts of the absorber's index of refraction, respectively) have been demonstrated to mitigate mask 3D effects, which become increasingly important for high NA EUVL. The low refractive index of these materials enables higher image contrast and a more efficient use of light. The optimum refractive index n, extinction k and thickness of the absorber cannot be found by standard thin film considerations. It depends on mask tonality, mask design options, the illumination used and exposure strategy. An industry wide consensus on few appropriate mask blanks with a broad coverage of applications is required to identify the most cost-efficient solutions.

ILT (inverse lithography technology) is expected to help to extend low-NA EUVL technologies by enabling wider process window and thus better yield. Its utilization continues to be an extension of existing imaging and material technologies, while development on the next generation solutions matures in both capability and costs.

Increasing use of curvilinear mask features is expected in both leading-edge photomasks (where maximizing process windows via ILT and other reticle enhancements are the drivers) and mid-range photomasks (as photonics and other non-IC applications

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increase in the semiconductor manufacturing space). This complexity is driving improvements in data representation, notably with the MULTIGON extension to the SEMI P39 standard,[5] to reduce the impact of data volume. It is also driving the wider use of multi-beam mask writers, using a raster-based exposure method, to control mask write times. However, further improvements will be required in all aspects of mask manufacturing, including EDA software (to handle sizing and other operations natively for mask data preparation), metrology, inspection, and repair. These improvements will also prove necessary in the mid-range space, as more photonic content will drive the use of curvilinear data formats.

4.3. Stochastics

In EUV lithography there are stochastic failures, a class of defects not typically seen in optical lithography. However, they occur at the quantum level and have become significant for EUV lithography. They are caused by photon shot-noise (fluctuations in the number of photons), the positions of photon absorption, the trajectories of photon-generated electrons and in the positions and occurrences of the chemical reactions that change resist solubility. Stochastic defects can be manifested as line bridges, broken lines, bridged contact holes, or missing contact holes. Their frequency needs to be very low, so fabs do extensive testing to characterize their frequency. Processes used in high volume manufacturing (HVM) have sufficiently low levels of stochastic defects to ensure adequate yield.

The stochastic defect rate is very sensitive to pitch and feature size. In some cases, this has already necessitated the use of multiple patterning. The steep dependence of stochastic failure as a function of pitch suggests that shrinking pitches for future nodes and still meeting defectivity targets will be a challenge. There is risk that the smaller dimensions projected for future nodes in the IRDS roadmap will have stochastic failures limiting yield. In addition, at certain frequencies, stochastic failures could affect the defect-limited yield measured in a fab but be too low in frequency to be detected in product development.

Defectivity is expected to be a particular challenge in the long term. The 2024 More Moore roadmap projects that defects per square inch will have to decrease by over 50% per node after 2031 when stacking of devices is used to increase device density. Any source of defects that cannot be reduced by process adjustments will be a major challenge. Defectivity could limit the size of chips that can be made at reasonable yield.

4.4. Light Source Challenges

4.4.1. Introduction

Light source innovation is critical for both lithography and metrology/inspection.[6] EUV light source innovation and industrialization has opened new frontiers in advanced device high-volume manufacturing and subsequently in metrology and inspection applications. Cost and capability remain the two dominant drivers for light source innovation. Even when numerical aperture and optical column transmission improve, source power improvement is still beneficial for lithography throughput, yield, and performance. Other improvements such as control over light qualities such as coherence, polarization, pulse duration, spectral bandwidth and purity could also be beneficial. The use of a wavelength shorter than 13.5 nm is still under consideration and will require radically new light sources.

The industry's transition from optical to EUV lithography highlights the steep adoption curve required for any transformative light source technology to enter the fab. While especially true for lithography moving below 13.5 nm, metrology and inspection toolsets leveraging VUV, soft x-ray, or hard x-ray sources also have a steep adoption curve and will require across industry support to provide economically feasible capability.

4.4.2. Lithography Light Sources

Laser-produced plasma (LPP) EUV light sources are currently used routinely for EUV lithography in HVM with high-reliability and availability approaching 97% in the field.[7] An EUV power of >500W at intermediate focus will be enabled on the next generation of tools, and 740 W has been achieved in the lab. Efforts are underway to increase this to ≥1 kW. LPP power scaling is achieved through improved target formation from pre-pulse conditions, higher repetition rate, and increased drive laser power. Increased repetition rate requires faster droplet speed to maintain spacing for the Sn target material, and repetition rates of 100 kHz are being explored. Using a separate pre-pulse laser for target formation allows higher usable main pulse powers from the drive laser. As output power increases, so does the amount of electricity required to produce the EUV light. There are several options for reducing the amount of electricity consumed. Infrared lasers with wavelengths shorter than that produced by the CO_2 drive lasers used currently may provide better efficiency in converting electricity to infrared light and for converting infrared light to EUV light. Improving the efficiency of the scanner optical train will also reduce the total electrical power required to produce each wafer.[8]

Several alternative light source architectures have been proposed in the past decade as either LPP alternatives, replacements, or future enablers, including free-electron lasers,[9] [10] steady-state microbunching synchrotrons (SSMB),[11] compact regenerative

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amplifiers,[12] and traditional synchrotrons.[13] Among these, only large-scale accelerator facilities are still being pursued in earnest. Both free-electron lasers[14] [15] and synchrotrons[16] [17] have a sufficient base technology maturity to potentially be accessible to the semiconductor industries. However, a synchrotron-based lithography light source at sufficient power to match LPP performance still requires additional research and technology demonstration.[18] Previous work on synchrotrons had indicated sufficient power for HVM applications could not be extracted; however, recent theoretical work has highlight the potential for improving the output power through both aggressive extraction efficiencies and beam cooling.[19] Similar to SSMB, these novel synchrotron configurations require experimental testing.

Free-electron lasers (FELs) offer unique capabilities in EUV power, light quality, and flexibility (e.g., wavelength and point application tunability).[20] FELs offer the ability to provide kWs of EUV power at availability approaching 100% without out-of-band radiation and at narrow bandwidths (which will improve optical column transmission), for less than a sixth of the electrical requirement of LPP sources. An FEL is capable of powering multiple lithography tools simultaneously and will have to do so in order to be cost effective.[21] FELs intrinsically produce polarized light that is programmable based on the design of the emission architecture, allowing polarized light delivery to an application without significant losses.[22] These potential improvements over laser pulsed plasma systems drive the present consideration of FELs as an EUV light source; however, the necessary paradigm shift to a "utility-scale" light source architecture is sufficiently disruptive to necessitate broad consensus on its development and the associated ecosystem / infrastructure changes.

With the advent of high-NA EUV lithography as well as the current investigation into EUV hyper-NA scaling, the potential of developing short wavelength (6.x nm, BEUV)) plasma-based sources bears consideration once again. Gadolinium,[23] terbium,[24] xenon,[25] neon,[26] other Pt-like ions, and alloys[27] are the main candidates for plasma fuel material, which may be intractable based on toxicity, cost or implementation challenges. The industry must make an ecosystem development decision in the next few years as to whether hyper-NA 13.5 nm or BEUV is preferred, based on economics of source technology development and potential patterning benefits (e.g., etch process latitude, resolution (λ/NA), and DOF (λ/NA^2)) correlated to device architecture. Lastly, it is essential that any innovation in wavelength or platform maintain or improve performance in edge-placement error (EPE) and throughput. The decision will depend both on the estimated capabilities of improved 13.5 nm imaging and on the estimated source capabilities (both plasma and FEL based), lithographic performance and throughput at 6.xnm wavelength.

4.4.3. Metrology and Inspection Light Source Outlook

Recent advancements in semiconductor devices have led to increasingly complex 3D architectures, therefore necessitating a diverse range of inspection, metrology, and materials analysis techniques. These techniques utilize various light sources spanning from hard X-rays with wavelengths as short as ~0.01 nm to terahertz (THz) radiation with wavelengths as long as ~1 mm suitable for different applications as described in Fig.1. While each of these applications can be served by a specialized source, it is also possible that all applications are serviced by a single source as noted above.

In the field of semiconductor metrology and inspection equipment, X-ray sources generated by accelerated electron beams colliding with metals are commonly employed. Depending on the speed, energy, and type of metal, these sources produce continuous spectra known as Bremsstrahlung and discrete spectra known as Characteristic X-rays.

EUV light sources are used for reticle inspection and are typically implemented through two methods. Firstly, coherent EUV light can be obtained using high harmonic generation (HHG) from laser sources, providing advantages such as compact size and wavelength tunability despite lower source powers ($\mu W \rightarrow mW$). These sources have advanced in industrialization with respect to the source architecture beyond academic designs, and similarly, industrialized drive lasers (both fiber and solid-state) have evolved to a level of reliable compatible with exploratory application development. To date, there has not been an effective harnessing of the coherent nature of this light source to offset the greatly increased photon flux offered by incoherent alternative sources. As drive lasers and coherent EUV source architectures continue to evolve in the coming years, it is possible that these sources transition from the lab to the fab. However, reliabilities of >90% must be achieved at stable output powers at relevant wavelengths with sufficient tunability to offer capability over photon flux. Alternatively, incoherent EUV radiation can be produced by generating high-energy laser produced plasma (LPP) using lasers and fuel material (tin, copper, xenon, etc…), offering high output power but posing challenges in installation, operation and cost. These sources continue to scale in output power and usable brightness, however in-fab operational efficiency requires >1kW/mm sr as measured at the source (400W / mm sr realized today).[28]

The sensitivity of scatterometry measurements peaks in the VUV spectrum, offering the potential for chemically specific probing of three-dimensional structures.[29] Industrialization of VUV sources therefore offers a means to improve process control sensitivity in the fab outside of computational algorithms (but would require introducing a vacuum load-lock to such toolsets and applications). Vacuum ultraviolet light sources are poorly industrialized currently, with only accelerator-based light source and

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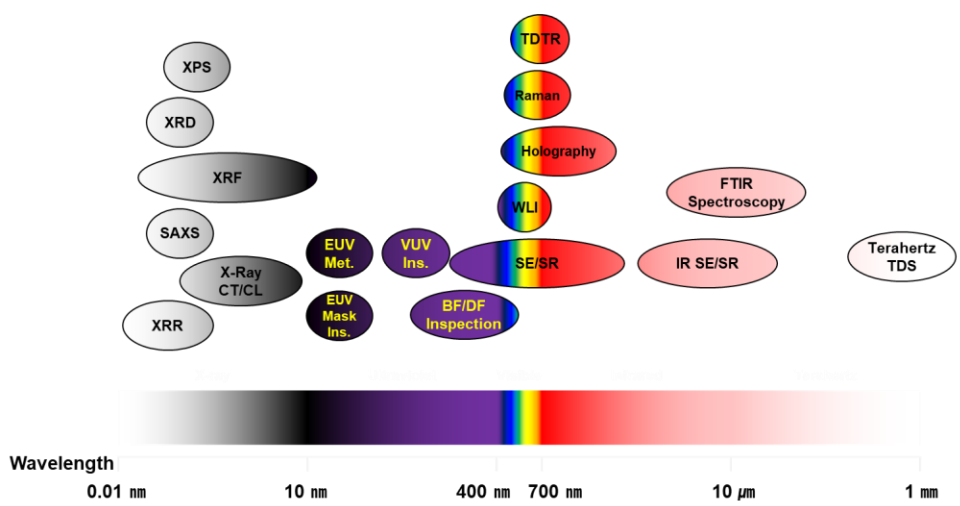
few table-top systems available (and none at the power and performance required for metrology and inspection applications).[30] [31] Introducing VUV scatterometry for CFETs / LGAA and especially 3DVLSI would be an enabling technology.

Ultraviolet (UV) radiation, including Deep UV (DUV), alongside visible and infrared regions, is extensively utilized in semiconductor metrology and inspection technology. High-end bright-field (BF) inspection equipment, crucial for detecting very small defects around ~10 nm, employs UV radiation ranging from 200 nm to around 450 nm wavelengths. BF inspection equipment technology that has been extended to EUV, and now the Vacuum UV (~100 nm) wavelength range has emerged as a future technology candidate. The higher available powers, robust mirror technology, and index contrast in this regime make it an attractive region. Optical techniques like spectroscopic ellipsometry (SE) and spectroscopic reflectometry (SR) also utilize wavelengths from 170 nm DUV to 2000 nm NIR for optical critical dimension (CD) measurements.

Visible light, being familiar and accessible for scientific research has fostered the development of diverse technologies. Techniques such as BF inspection equipment for detecting macro defects ranging from hundreds of nanometers to micrometers in size, White Light Interferometry (WLI) for rapid wafer surface profiling, Holography for obtaining both amplitude and phase information, Time-Domain Thermal Reflectometry (TDTR) for detecting internal defects based on reflectivity changes over time due to heat flow, and Raman Spectroscopy for analyzing material properties based on signal changes by Raman scattering, are among techniques predominantly utilizing visible light. From UV to visible and into the infrared regions, lamp and laser light sources are commonly used depending on the specific requirements. Gas discharge lamps like Arc Lamps, Plasma Lighting Systems (PLS) utilizing plasma generated by electromagnetic waves in inert gas, and Laser Produced Plasma (LPP) as seen in EUV and visible sources are suitable for applications requiring broad spectrum illumination. For applications requiring intense monochromatic light, multichannel laser sources offer versatility depending on factors such as cavity structure and gain medium.

These days, technologies utilizing longer wavelength ranges from mid-infrared (~5 µm) to THz (~1 mm) have been increasingly applied in semiconductor manufacturing. Fourier Transform Infrared (FTIR) spectroscopy for molecular structure and properties analysis, IR SE/SR for thickness measurement in deep structures like 3D NAND, and THz Time-Domain Spectroscopy (THz TDS) for electrical property characterization are prominent examples. Mid-infrared light sources typically rely on thermal radiation principles, quantum cascade lasers, or solid-state/diode lasers, while THz light sources often utilize techniques like photoconductive antennas or optical rectification, both requiring femtosecond pulse lasers as pump for generating pulsed THz waves.



(Ins. = Inspection, Met. = Metrology, BF = Bright field, DF = Dark field, WLI = White light interferometry, SE = Spectroscopic ellipsometry, SR = Spectroscopic reflectometry, TDS = time domain spectroscopy, CT/CL = computed Tomography/computed laminography)

Figure LITH-1 Various metrology and inspection techniques for different wavelengths.

4.4.4. Summary

Light sources are at the core of device innovation and the economics of high-volume manufacturing. Within the decade, EUV lithography light sources will scale to a kilowatt of power at the intermediate focus, and the patterning roadmap must describe the requirements for light qualities outside of raw power. Similarly, as device architecture becomes both denser and three-dimensional, new metrology and inspection applications will come to fruition, driving power, brightness and wavelength

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requirements to sustain process development and manufacturing cycle times Lastly, industry consensus for "utility scale" light sources as both capability and performance driver must be reached on the topic of accelerator-based light sources if they are to be implemented.

4.5. Potential Solutions

The table below shows Potential Solutions incorporating all the factors described above. High NA EUV is expected to be used in early manufacturing in 2027, potentially being used in both single and double patterning. Hyper NA could be in use as early as 2033 if this option is pursued. A shorter wavelength could be implemented in 2035 if that option is pursued. Many of these nodes will probably also require patterning and materials innovations as described below.

Table LITH-4 Potential Solutions for Leading-Edge Logic Lithography

	2025	2027	2029	2031	2033	2035 and beyond
Logic node	"2 nm"	"1.4 nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq" and beyond
Node	G45M22	G42M20	G42M20	G42M18	G40M16/T2	G40M14/T4 and beyond
Minimum ½-pitch	11 nm	11 nm	10 nm	9 nm	8 nm	7 nm
Primary options for logic	EUV 0.33.NA multiple patterning	EUV 0.55.NA single patterning EUV 0.55.NA multiple patterning EUV 0.33.NA multiple patterning	EUV 0.55.NA single patterning EUV 0.55.NA multiple patterning EUV 0.33.NA multiple patterning	EUV 0.55.NA single patterning EUV 0.55.NA multiple patterning EUV 0.33.NA multiple patterning	EUV 0.55.NA single patterning EUV 0.55.NA multiple patterning EUV >=0.75NA single patterning	0.55.NA multiple patterning EUV >=0.75NA single patterning Beyond EUVL (λ=6.X nm)

5. Patterning Challenges

5.1. Introduction

As described above, pitch scaling will continue to scale at least until a minimum pitch of 14 nm is reached, at which point scaling by stacking is introduced. Pitch scaling may be slowing down due to DTCO innovations and advanced packaging, but industry demand for more advanced resolution is still very strong with special focus on improving cost efficiency of the patterning and minimizing line edge roughness (LER) and edge placement error (EPE). To achieve this, targets to focus on are:

- Progressive replacement of SALELE (self aligned litho etch litho etch) levels with single print
- Minimize lithography passes, either by improved resolution or by alternative technologies (i.e. pattern reshaping)
- EUV capability extension
- Dose reduction by using resists with higher optical absorption, such as metal-oxide resists (MORs)

These targets require both new lithographic materials and specifically designed capabilities to add to the patterning ecosystem.

5.2. Patterning Materials

Patterning materials remain a key challenge for extending scaling to high NA EUV. Along with photoresist development, co-optimization of the underlayer and patterning stack are necessary for improved patterning performance. Chemically amplified resists (CARs) have been the workhorse for both single and double patterning options for critical device and metal levels at 0.33

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NA, but they have shown limited performance at pitches ≤20 nm for single expose L/S patterning with high NA EUV[32] and need further efforts to improve performance for high resolution.

Metal oxide resists are the biggest competitor to CAR, with their higher photon absorption that could translate to lower doses,.[33] [34] These materials have started to be used in production. While some potential applications in 0.33 NA double patterning can be foreseen, the differentiator for MOR will be with high NA (0.55) EUV insertion. With this higher resolution capability, MOR will be expected to resolve two dimensional (tip-to-tip) structures and give the advantage of reducing the number of EUV masks by enabling single exposure. Continuous improvement of MOR platforms will be expected along with increasing quantum yield to improve roughness and defectivity without compromising on dose. As the patterning stack gets thinner in order to maintain printable aspect ratios despite shrinking pitch, the resist underlayer interface tuning will be key. Both spin on and deposited underlayers are being optimized with respect to stochastic defects and pattern transfer.[35] [36] [37]

Track based edge bead removal and backside/bevel clean options (which have become standard processes to address metal contamination) will continue to evolve. Understanding the effects of the environment on their stability remains a significant focus for MOR.[38] CD stability control will be critical to enable stitching in high NA EUV lithography. New types of process flows with resist coat, exposure, bake and develop steps being developed.[39] Characterizing the effect of these new methods and their impact on resist performance is a near term industry focus.[40]

Wet and dry development are both being considered for MOR. While wet development uses the existing equipment eco-system, new hardware improvements are needed to optimize the performance for MOR.[41] A new coater/developer-based development method was introduced recently for MOR development for improving resolution, roughness, sensitivity, pattern collapse and defectivity.[42] [43] Dry deposited and dry developed resist is a new candidate in the industry that is demonstrating its capability through defectivity and yield validation.[44] Dry development has the advantage of enabling higher aspect ratio without pattern collapse. The dry technology enables in situ process tuning. Tuning of underlayer and pattern transfer techniques for MOR are also significant areas of development that can enhance its performance.[45]

While bright field L/S and pillar masks have become HVM ready, contact hole masks still suffer from mask defect issues. Hence, despite the resolution benefits of MOR, CAR remains the front up approach for this application.[46]

5.3. Patterning Integration/Ecosystem

Several methods are under evaluation to enable improved patterning performance and cost structure:

- Angled etch for pattern reshaping
- EUV rectification via DSA
- Advanced Underlayers
- Nano Imprinting Lithography (NIL) replacing EUV Litho for selected levels

Angled etch is a reactive ion etch (RIE) technique where the electric field used to create anisotropic etching is not normal to the substrate. Instead the electric field is at a steep angle to the vertical. Usually, the resist pattern is transferred into a substrate using a normal vertical RIE and the etched substrate undergoes the angled etch Because of the steep angle one side of an opening will be etched much faster than the other sides. One application of this is to improve tip to tip distance of trenches. The trenches are etched with the electric field directing the etch first at one end of the trenches and then at the other end. The benefit is that sub-lithographic tip-to-tip (T2T) distances can be produced without requiring a dedicated mask for line cuts, thus saving one exposure step. Such a process can also be used for microbridge removal. Of course, the pattern design has to have all the trenches running in the same direction to accommodate this process. Pattern elongations up to 20nm and defectivity reduction of 10X has been demonstrated.[47]

DSA (directed self-assembly) pattern rectification is potentially a way to get rid of stochastic defects and reduce LER while exposing at a relatively low EUV dose. For example, line to line bridges caused by rough line edges can be healed by etching an underlayer with the original resist pattern and using the etched underlayer as a template for DSA. The final pattern will have LER and pitch defined by the block copolymer. Because the LER of lines generated by DSA has a fundamentally different source than lines in resist, the DSA pattern can have much better roughness than the original resist pattern. A high speed resist can be used for such a process, thus improving EUV exposure throughput and reducing costs. DSA generates a limited range of patterns, so a special process flow is used to generate flexible patterns. This involves an additional exposure to trim the line and space patterns. It was reported that this sort of process enabled 18nm pitch metal line and space patterns that were undoable with a conventional process due to high variability.[48] LERs as low as 1nm at 24nm pitch were reported.[49] The actual cost of ownership benefit of using a more complicated process to give a rectified pattern based on a fast resist needs to be estimated case by case.

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DSA also has the potential for pitch multiplication. Work has been reported for multiplying the pitch of holes for memory[50] and for CMOS image sensor applications.[51] It is not clear if these pitch multiplication processes meet all the requirements for mass production.

Advanced EUV underlayers are another widely explored approach to improve the cost efficiency of patterning steps.[52] [53] They can provide either enhanced adhesion for CARs that enables smaller pitch without patterning defects, or enable lower dose for EUV resist. Underlayers provide a clear dose benefit in practice ongoing research, but mechanism of enabling lower dose is not yet understood and is the subject of ongoing research.[54]

Nanoimprint Lithography (NIL) has received a lot of attention recently as an alternative patterning method. For example, it might be applied as an EUV replacement for selected levels like SADP loop cuts or for word line cuts in DRAM flows.. NIL technology is rapidly improving its performances in terms of defectivity, throughput and alignment,[55] However it does not yet match EUV Lithography in some parameters. Defects and overlay are key performance parameters for this technique.

6. Design Technology Co-Optimization

As described in the 2023 IRDS Lithography update, patterning schemes such as fully self-aligned vias (FSAV) and self aligned contact over gate (SAC for COAG) enable better edge placement error for a given overlay and CD control capability. Their use will continue and will need improved materials as described in that report.[56]

In addition to these methods, this year backside metallization is in the roadmap and will have a significant impact on lithography. As detailed in the IRDS 2024 More Moore roadmap, various functions will move to the back of the chip instead of the front, starting with power distribution and some I/O in 2025 and continuing with clock and bus functions at a later date and eventually having active devices on the back side. Moving power routing to the back side of the chip enables substantial reduction in the area of the unit cell without shrinking any critical dimensions. Basically this makes the full 3D unit cell structure, top and bottom, more three dimensional, thus reducing the two dimensional area required. But this change entails patterning on the backside of the wafer at a pitch that matches pitches on the front side and with overlay sufficient to line up to through vias coming from the front of the chip. In 2025, this will require ArF lithography to print lines and spaces on a 180 nm pitch and via/contacts on a 120 nm pitch. In 2027 the required pitches will shrink to 32 nm and will require EUV lithography. Pitches continue to shrink from node to node until, in 2039, the minimum backside pitch is 16nm, almost as small as the minimum projected frontside pitch of 14nm. The table below shows the projected backside lithography requirements for the next logic three nodes.

Table LITH-5 Projected Backside Lithography Requirements

YEAR OF PRODUCTION	2025	2027	2029	2031	2033	2035	2037	2039
Number of Backside Metals	4	4	5	5	6	6	6	6
Backplane intereconnect 1/2 Pitch (nm)	120	32	27	24	22	16	16	16
First Backplane Contact (Via?) Pitch (nm)	180	32	27	24	22	19	19	23
Backplane Functions	Power distribution IO	Power distribution, IO, Clock Bus	Power distribution, IO, Clock Bus	Power distribution, IO, Clock Bus	Power distribution, IO, Clock Bus	Power distribution, IO, Clock Bus	Power distribution, IO, Clock Bus	Power distribution, IO, Clock Bus

7. Packaging Lithography

The current IRDS roadmaps show that increases in system performance will now be driven not only by shrinking of dimensions and more compact unit cell designs, but also by delivering modules that integrate multiple chips or chiplets. For example, for GPUs, it has been said that "The continuation of the trend of increasing transistor count [for GPUs] will require multiple chips, interconnected with 2.5D or 3D integration, to perform the computation...We forecast that within a decade a multichiplet GPU will have more than 1 trillion transistors".[57] Increased system integration of this sort will drive improved packaging lithography, including both better resolution and overlay. Many details of packaging can be found in the IRDS packaging roadmap. Here we only consider the lithographic issues associated with improved packaging.

Integration of many chips can be accomplished using interposers. Interposers are used by TSMC (CoWoS) and others to integrate high-performance chips for AI and other applications. Today, such interposers are limited to 3.3 reticle fields. However, large-format interposers will soon scale to full wafers[58]. Interconnection density between chiplets will also improve[59]. Adaptable patterning will enable tighter pitches and elimination of bonding pads, bringing benefits in chip-chip communication bandwidth and energy efficiency. We considered that increases in device density in logic chips, increases in required data rates and decreases

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in minimum line pitches will drive proportional increases in packaging bump density and redistribution line density. The table below shows packaging lithography targets are expected to improve.

Table LITH-6 Packaging Lithography Targets

Silicon interposer connections [B]	2024	2025	2027	2029	2031	2033	2035	2037	2039
Packaging density shrink factor [A}	1.00	0.95	0.81	0.74	0.62	0.59	0.46	0.39	0.30
Max. substrate size	3.3 reticle fields (2,831 mm^2)	3.3 reticle fields (2,831 mm^2)	Full wafer	Full wafer	Full wafer	Full wafer	Full wafer	Full wafer	Full wafer
RDL pitch in backend process (um)	3.00	2.86	2.44	2.22	1.87	1.77	1.38	1.16	0.90
Solder bump pitch (um)	10	9.5	8.1	7.4	6.2	5.9	4.6	3.9	3.0
Max. x,y placement error between chip pad and interconnect line (um)	2.0	1.9	1.6	1.5	1.2	1.2	0.9	0.8	0.6
Chip to Wafer stacking connections [C]									
Hybrid bonding pad pitch (microns)	9.0	9.0	6.0	3.0	2.5	2.4	1.9	1.6	1.2

Current large field projection exposure systems are specified to print down to 2 µm lines and spaces (2000 nm) or a little less and have throughputs on the order of 60 wph. The flatness of the substrates will affect the suitability of these tools. Silicon wafers are certainly flat enough, but other types of substrates may not be. Optical laser direct write systems provide maskless lithography that can resolve down to less than 2 µm and may have higher tolerance for warped or non-flat substrates. However, throughputs currently are on the order of an hour or more for 200 mm substrates and bigger. Optical laser direct write throughputs are slower when handling non-flat substrates. Multibeam ebeam tools are becoming available and have a very wide range of depth of focus and resolutions. Multibeam ebeam systems handle non-flat substrates with no change in throughput, provide resolution from microns down well below 100 nm lines and spaces and provide much higher throughputs than other ebeam systems. Their throughput compared to optical systems is pattern dependent and is generally in the range of 2 wph to over 25 wph per process module (6 wph to over 7 5 wph per system). A new sort of optical direct write system using 248 nm lithography is under development has been reported and may be available in 2025. This tool has excellent resolution of 110 nm, but low depth of focus and modest throughput of 2 wph for 300 mm wafers. An alternative to large field exposure is to use conventional steppers for exposure and then stitch fields together for large field sizes, but this is probably only applicable to interposers and other substrates made from silicon wafers. Patterning of large silicon wafers is likely to be a significant cost challenge in the future and patterning on substrates that are less flat could also be a significant technical challenge.

Direct write using both optical methods[60] and direct write e-beam methods[61] are being pursued. High-resolution adaptable patterning, enabled through direct write e-beam, enables interconnect density and bandwidth approaching metal interconnect layers in monolithic chips. The table below shows projected direct write capabilities by year.

Table LITH-7 Projected Direct Write Capabilities by Year

Year of production	2024	2027	2030	2033	2036
Line-width (nm)	150	72	36	22	16
Pitch (nm)	300	144	72	44	32
Overlay error (nm)	45	21	10	6.6	4.8

8. SUSTAINABILITY

Energy usage and other resource consumption is more and more a concern in the world. The semiconductor industry is already sensitive to the cost per transistor along with the computational gain for each successive device node. Now an additional question is whether the increased device density and chip performance is outpacing the use of energy, water, and other resources used by fabs. For lithography, there are two different issues when it comes to sustainability. One is the consumption of resources, such as electrical power and water, and one is the use of materials that may have deleterious environmental effects. These issues are discussed in the Environment, Safety, Health and Sustainability (EHS) IRDS reports. [62] The EHS reports cover these issues for the whole semiconductor production process. Here we only discuss issues specific to lithography. There is active work underway on alternative chemicals to replace fluorine containing ones and on new EUV light sources that potentially have higher efficiency. This is promising for improving the sustainability of lithography processes.

One of these issues is energy usage. As described above, current EUV light sources are inefficient at converting electrical power into usable EUV light. With the current approach using laser-produced plasmas generated by CO_2 lasers, 1.3-1.8 MW of electricity will be required to provide 1 kW of EUV light to a single EUV exposure tool. Despite the reduction of process steps

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involved in using EUV instead of multiple patterning, EUV energy consumption is a concern. Possibilities for reducing energy consumption are discussed in the source section above.

Another opportunity for reducing overall energy consumption in EUV lithography is through hydrogen recycling.[63] Hydrogen is used in EUV systems to remove carbon contamination and reduce oxidation of mirror and masks surfaces, as well as reduce mirror contamination from the tin used in laser-produced plasma EUV light sources. Per exposure tool, hydrogen consumption is ~600 standard liters per minute (SLM), so recycling can be very beneficial.

The other issues concerns certain chemicals that contain fluorine, particularly PFAS (Per- and Poly- FluoroAlkyl Substances) related materials. PFAS-containing chemicals are currently used to provide certain functionality in lithographic materials, such as high-performance photoacid generators for chemically amplified resists, top anti-reflection coatings (TARCs), top-coats for immersion lithography, and surfactants and surface property modification additives for photoresist to prevent pattern collapse. Because of their persistence in the environment, and possible harmful health effects, many governments are taking steps to limit PFAS releases into the environment.

Although the actual amount of PFAS used in lithography is relatively small and is estimated to be between 0.07-0.10% of the worldwide use, addressing the issue of PFAS for lithography applications is part of the worldwide effort to eliminate PFAS emissions into the environment.[64] There is ongoing research into replacements for PFAS materials, although the unique qualities of PFAS make replacements difficult. Some potential PAG replacements for i-line and KrF resists have been described recently.[65] [66] However, the unique properties of fluorine in enhancing acid strength and in enabling low refractive index polymers make replacing certain PAGs and replacing polymers for TARCs and top-coats particularly difficult. The long lead time for qualifying new materials in semiconductor processes along with the time it takes for new materials R&D mean that PFAS replacement in lithography uses would benefit from a long exemption time.

Depending on the material, control of waste streams may offer better opportunities to reduce environmental release in the short term.[67] Organic waste streams account for about 50% of the PFAS used in lithography, and this waste is collected and destroyed. Aqueous waste accounts for the other 50%, primarily from TARCs, and here there exists opportunities for significant reduction though development bowl waste collection and proper disposal.

9. Long Term Challenges

One long term challenge is defectivity. Defectivity will have to improve significantly when devices start being built in tiers and feature sizes shrink. Required defect rates are described in the More Moore roadmap. Since stochastic defects increase exponentially with shrinking feature size, this is a significant challenge. As described above, another challenge is determining what is the best path beyond 0.55 NA EUV. Another issue is how to make larger integrated circuits (ICs) than can be printed in one exposure with high NA EUV. As described above, there are different options for addressing larger ICs but they are all expensive. Stitching requires exposing a wafer with two or more reticles, larger mask sizes require mask and tool infrastructure investments and using small ICs connected with high performance packaging requires expensive packaging processes. Packaging lithography is also expected to be a challenge. Large substrates often require high DOF but exposure methods that provide high DOF and smaller pitches at the same time are typically quite slow and thus expensive. What is more, the tight overlay requirements required for printing and assembling packages will add substantially to the equipment cost. Finally, although most half pitches in the requirements table are achievable with 0.55 NA tooling, there is not a demonstrated method for the 7 nm half pitches projected in the later years of the roadmap.

Table LITH-8 Long-term Challenges

Reduced stochastics and defectivity
Yield of stacked devices
What comes after 0.55NA EUV
Large area devices or equivalent of larger exposure field size
Cost of packaging lithography
Manufacturing methods for 7 nm half pitch

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10. ACRONYMS/ABBREVIATIONS

Term	Definition
ArF	Argon fluoride
ASD	Area selective atomic layer deposition
BEUV	Beyond EUV
CARS	Chemically amplified resists
CD	Critical dimension
CDU	Critical dimension uniformity
CMOS	Complementary metal oxide semiconductor
COAG	Contact over active gate
CoO	Cost of ownership
CVD	Chemical vapor deposition
DOD	Dielectric on dielectric
DOM	Dielectric on metal
DRAM	Dynamic random access memory
DSA	Directed self-assembly
DTCO	Design-technology co-optimization
EPE	Edge replacement error
ESH&S	Environment, safety, health and sustainability
EUV	Extreme ultraviolet
FEL	Free electron laser
FET	Field effect transistor
FinFET	Fin field-effect transistor
FSAV	Fully self-aligned vias
GAA	Gate all around
GPU	Graphic processing unit
HM	Hard mask
HVM	High volume manufacturing
I/O	Input/output
IC	Integrated circuit
IEEE	Institute of Electrical and Electronic Engineers
IEEE-SA	IEEE Standards Association
IFT	International focus team
IMEC	Interuniversity Microelectronics Centre
ILD	Interlevel dielectric
iNEMI	International Electronics Manufacturing Initiative
I/O	Input/output
IRC	International roadmap committee
IRDS	International Roadmap for Devices and Systems
ISSCC	International Solid-State Circuits Conference
ITRS	International Technology Roadmap for Semiconductors
KrF	Krypton fluoride
LCDU	Local CD uniformity

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Term	Definition
LER	Line edge roughness
LGAA	Lateral gate all around
LPP	Laser-produced plasma
LWR	Line width roughness
MEMS	Micro-electro mechanical system
MRAM	Magnetic RAM
MOR	Metal oxide resist
MOS	Metal oxide semiconductor
MOSFET	Metal oxide semiconductor field effect transistor
MPU	Microprocessor unit
MRAM	Magnetic RAM
Mx	Tight-pitch routing metal interconnect
NA	Numerical aperture
NAND	A logic gate (NOT-AND) that produces an output that is false only if all its inputs are true
NIL	Nanoimprint lithography
OPC	Optical proximity correction
ORSC	Overall roadmap system characteristics
ORTC	Overall roadmap technology characteristics
OSC	Outside System Connectivity
PEALD	Plasma enhanced atomic layer deposition
PVD	Plasma vapor deposition
PTD	Positive tone wet development
RIE	Reactive ion etch
SAB	Self-aligned blocks
SAC	Self-aligned contacts
S/D	Source/drain
SDRJ	System Device Roadmap Committee of Japan
SEM	Scanning electron microscope
VGAA	Vertical gate all around
VLSI	Very large scale integration

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